



Atrazine Transport in Irrigated Heavy- and Coarse-Textured Soils, Part II: simulations with the Root Zone Water Quality Model

A. S. Azevedo¹; R. S. Kanwar²; L. S. Pereira¹

¹Centro de Estudos de Engenharia Rural, Instituto Superior de Agronomia, UTL, 1399 Lisboa Codex, Portugal;
e-mail of corresponding author: nop58971@mail.telepac.pt

²Department of Agricultural and Biosystems Engineering, Iowa State University, Ames, IA 50011, USA;
e-mail: rskanwar@iastate.edu

(Received 19 May 1999; accepted in revised form 24 February 2000)

The hydrological and the pesticide component of the root zone water quality model (RZWQM) were calibrated and evaluated for two different Portuguese soil types: silty loam and sandy soils, under irrigated conditions. The RZWQM is a one-dimensional, process-based model, that simulates the physical, chemical and biological processes in the root zone, and can also accurately simulate macropore flow. The model was calibrated using the 1997 data and tested using 1996 data (soil water content and atrazine concentration in the soil) collected at an Experimental Station located at the Sorraia valley, near Coruche, Portugal, during the summers of 1996 and 1997. Data on plant growth was also available for 1997.

The RZWQM is effective in simulating plant growth parameters. For both soil types (silty loam and sandy soils) model predictions on soil water content were in agreement with the observed values. During the calibration year, for most of the irrigation events, model performance was good. Regarding atrazine concentrations, model predictions were in the same range of the observed values. These results indicate that RZWQM can be used, when properly calibrated, to simulate the movement of water and atrazine through the soil profile, for Portuguese conditions.

© 2000 Silsoe Research Institute

1. Introduction

The degradation of natural habitats and the loss of biodiversity for plant and animal life has been a concern in the industrialized countries. The decrease or disappearance of many animal and plant species provided the warning in some parts of the world. Agriculture has been identified as a major contributor for non-point source pollution. In the past decade, the use of agrochemicals has greatly increased, in order to provide an adequate food supply for the growing population. However, these chemicals are responsible for surface and groundwater contamination. One of the most detected chemicals in groundwater (Funari *et al.*, 1995; Hallberg, 1989; Stoltenberg *et al.*, 1990) is atrazine (2-chloro-4-ethylamino-6-isopropylamino-1,3,5 triazine). Atrazine is an herbicide, commonly used in maize (*Zea mays*, L.) fields. It is the role of the scientists to investigate and provide to farmers the adequate technology to use these chemicals with minimum impacts to the environment.

To investigate the fate and transport of agrochemicals in the environment, mathematical models can be used to simulate multiple scenarios and evaluate their impacts, instead of collecting large volumes of field data over a long period of time (Ma & Selim, 1997; Pereira & Cameira, 1999). However, to get accurate results, and ensure that models are reliable, these models should be tested and evaluated against field data.

The root zone water quality model (RZWQM) is a one-dimensional, process-based model, that simulates the physical, chemical and biological processes in the root zone. The model has been used to simulate water table fluctuations and tile drainage (Johnsen *et al.*, 1995), to predict subsurface drain-flows quantity and its quality (Azevedo *et al.*, 1997b; Singh *et al.*, 1995, 1996), to evaluate pesticide transport through the soil profile (Azevedo *et al.*, 1997a; Ahuja *et al.*, 1996), to assess impacts of management on crop yields, and on evapotranspiration rates (Farahani *et al.*, 1995; Nokes *et al.*, 1995), and to characterize macropore flow and its impacts on water

quality (Ahuja *et al.*, 1993). This model has been tested for Portuguese conditions, at the same experimental station, to simulate water and nitrate movement in soils (Cameira *et al.*, 1997, 1998). A recent review is presented by Ahuja *et al.* (1998). However, the RZWQM has not yet been evaluated to simulate pesticide transport for Portuguese soils.

The objectives of this study were to: (a) calibrate the hydrological and pesticide component of the RZWQM for Portuguese conditions, using field measured data from 1997; and (b) evaluate the RZWQM by predicting soil water content and soil atrazine concentrations for 1996. The performance of the model was tested using field data from Coruche, Portugal, for silty loam (heavy-textured) and sandy (coarse-textured) soils.

2. Model description

The root zone water quality model, RZWQM (USDA-ARS, 1992) is a process-based model that simulates the physical, chemical and biological processes in the root zone that are affected by management. This model is a one-dimensional (vertical in the soil profile) model, that simulates conditions at a representative point in a field. The following management practices are included in the RZWQM: planting and harvesting, tillage events, fertilizer applications, pesticide applications, mode of chemical application and irrigation, including fertigation and chemigation. Incorporating all these processes enables the RZWQM to simulate many of the processes that occur during a normal growing season, as well as the effects of the interactions between multiple management practices.

Infiltration is described by the Green and Ampt equation (Ahuja, 1983), soil matrix water movement is simulated with the Richards equation (Celia *et al.*, 1990), and the macropore flow is computed adopting the Poiseuille equation (Ahuja *et al.*, 1993). A detailed description of the hydrological, nutrient and plant growth processes is given in Azevedo *et al.* (1997b). Therefore, only a brief description on the pesticide processes is given in the following section. For more detailed information on the RZWQM model, the technical documentation and the user's manual of RZWQM should be consulted (USDA-ARS, 1992, 1995).

2.1. Pesticide processes

The RZWQM considers three different types of processes that affect the pesticide movement, transformation and adsorption-desorption processes. The movement processes considered in RZWQM are: volatilization, pes-

ticide leaching into deeper soil layers, pesticide upward movement, runoff and pesticide transport with water. To measure pesticide volatility the Henry's constant is used (Nash, 1988). In this model, pesticide leaching is calculated in the hydrology model and pesticide solution concentration. The diffusion processes incorporated in the present version of RZWQM only consider pesticide diffusion in and between micropores and mesopores, and from macropore walls (Ahuja & Webson, 1992). Pesticide loss in runoff considers both runoff loss of pesticide in the solid phase (considering the modified universal soil loss equation) and through runoff water.

Regarding transformation, the key dissipation processes in the RZWQM include biotransformation (aerobic and anaerobic degradation), chemical oxidation and reduction, and chemical hydrolysis, photodecomposition and polymerization. Also, pesticide washoff and pesticide dissipation as daughter products are also included in the pesticide dissipation processes in the RZWQM. The modelling approach proposed by Wagenet and Rao (1990) is adopted in this model.

To account for the adsorption (sorption)-desorption under non-equilibrium conditions, a two-stage adsorption model (also called fast and slow adsorption model) was adopted in the RZWQM. Pesticide adsorption is divided in two stages, each stage having its own adsorption and desorption coefficients. The first adsorption stage is immediate, and the second one is slower taking from hours to a few days for some pesticides to reach equilibrium. In the RZWQM the adsorption coefficients for this model are estimated from pesticide sorption on organic carbon K_{oc} , if measured data are not available, as happens in this study.

There are four options for pesticide dissipation modelling in the RZWQM: (1) lumped pesticide dissipation by a one-compartment model; (2) pesticide dissipation using a two-compartment model, when the necessary data are available; (3) pesticide dissipation for each individual process; and (4) dissipation through pesticide daughter product formation and subsequent dissipation. The first option was adopted in this study.

2.2. Model inputs

2.2.1. Meteorological data

Daily values of air temperature (maximum and minimum), wind speed, pan evaporation or short-wave radiation and relative humidity are required as input to the model. Daily climatic data collected at Coruche experimental station weather station was used for experimental plots which were within 500 m from the weather station. The model requires input of rainfall data as breakpoint rainfall data. As only daily rainfall data were available at

the experimental station, these data were converted into breakpoint rainfall data assuming a uniform rainfall distribution over an 11 h period. The distribution over this time will not allow runoff to occur, which is valid assumption for this experimental area because (1) the plots are flat; (2) those in the silty loam soil were laser levelled and banded for level basin irrigation; and (3) those in the sandy soil were furrowed and banded on both ends and have very high infiltration rate, near 40 cm/h. Data on rainfall and reference evapotranspiration for 1996 and 1997 are given in the first part of this study (Azevedo *et al.*, 2000).

2.2.2. Soil properties data

For each horizon, the model requires data regarding the soil physical properties, such as soil type, particle density, bulk density, percentage of sand, silt and clay. Several hydraulic properties, such as saturated hydraulic conductivity, effective porosity, and bubbling pressure can either be specified as input by the user or estimated by the model based on bulk density, texture, and soil water content at 1/3 bar. Data on selected physical properties for the soils at the experimental site are given in the first part of this study (Azevedo *et al.*, 2000).

2.2.3. Management and field operations

Dates on planting, harvesting, fertilizer and pesticide application, tillage operations and irrigation events are used as inputs to the model. For each tillage operation, the effective depth and tillage intensity have to be specified. Regarding irrigation, the user can choose between surface and sprinkler irrigation, either fixing the intervals for each event, specifying the dates, or a level of root zone water depletion. For these simulations, the specific dates for each irrigation event were given to the model. Regarding pesticide management, application control parameters such as number of pesticide applications, method of application and the amount of active ingredient applied, are needed as inputs to the model. Management operations are presented in the first part of this study (Azevedo *et al.*, 2000).

2.2.4. Pesticide parameters

Parameters such as half-life, solubility, vapour pressure, sorption constant for organic matter K_{oc} are required by the model. In this simulation study, half-life was the calibration parameter. First estimation of this parameter was obtained from field observations (Azevedo *et al.*, 2000).

3. Materials and methods

3.1. Observed data

The model was calibrated and evaluated using field data collected at the Antonio Teixeira's Experimental

Station, located at the Sorraia valley, near Coruche, Portugal, during the summers of 1996 and 1997. Two maize cropped plots were used, one located in a silty loam soil, of recent alluvial origin and the other a sandy soil, of ancient alluvial formation. Physical properties for the soils were measured for each soil layer in 1996. The silty loam soil plot was under surface irrigation (furrowed level-basin irrigation) during the summers of 1996 and 1997. The sandy soil plot was under surface irrigation during the summer of 1996 and under sprinkler irrigation during the summer of 1997. Atrazine was surface applied at a rate of 1.2 kg a.i./ha in the silty loam soil plot, and at a rate of 1.0 kg a.i./ha in the sandy soil plot, in 1996, two days before planting, in both plots. In 1997, atrazine was applied at the same rate, nine days before planting in the silty loam soil plot, and six days before planting in the sandy soil plot. Atrazine application was broadcast following the current farmers practice, which is one of the management options considered by the model. Complementary information on field experiments, including the dates of main operations, is given in the first part of this study (Azevedo *et al.*, 2000).

3.2. Calibration and evaluation procedure

To calibrate the RZWQM in this study, a three-step calibration procedure was used. First, the plant growth component of the model was calibrated to predict yields. In the second step, the hydrological component of the RZWQM was calibrated to adjust the soil moisture content. However, because the plant growth and the hydrological component are closely related, the plant growth component was again checked after this last calibration. Finally, the pesticide component was calibrated to predict pesticide dissipation through the soil profile, along the growing season. Data from 1997 were used to calibrate the model, because more data were available for this year. The model was evaluated using data from 1996.

3.2.1. Calibration of the plant growth component

To calibrate this model component for maize, the site-specific parameters (maximum nitrogen uptake rate in g/plant/day, proportion of photosynthate to respire, amount of biomass needed to obtain leaf area index of 1.0, age effect for propagules as a proportion of photosynthate, and age effect for seeds as a proportion of photosynthate) were adjusted to predict yields close to the observed yields, leaf area index (LAI) and total above ground biomass.

During 1997, LAI and dry weight were measured for the crop in each of the plots. Three plants were collected

Table 1
Calibrated parameters for the plant growth component of the model

Site specific parameters	Value
Maximum nitrogen uptake rate, g/plant/day	5.50
Proportion of photosynthate to respire	0.12
Amount of biomass needed for a leaf area index of 1.0, g	9.7*; 10.20†
Age effect for propagules as a proportion of photosynthate	0.85
Age effect for seeds as a proportion of photosynthate	0.52

*‘Capitan’ maize, in the silty loam soil. †‘Franca’ maize, in the sandy soil.

for each plot. Leaf area measurements were made using a leaf area meter. The value for LAI was obtained dividing the measured leaf area value by the area occupied by the plant. To obtain the dry weight of the sample, five plants from each plot were kept in the oven at 65°C until a constant weight was obtained (after about 4–5 days).

The values of the calibrated parameters are given in Table 1.

3.2.2. Calibration of the hydrological component

The soil water retention and the hydraulic conductivity curves defined by Brooks and Corey (1964) are used in the RZWQM with slight modifications (Ahuja & Web-son, 1992). The soil water retention curve is given by

$$\theta(h) = \theta_s - A_1/h, \quad 0 \leq h \leq h_{b1}/$$
 (1)

$$\theta(h) = \theta_r + B/h^{-A_2}, \quad h \geq h_{b1}/$$
 (2)

with

$$B = (\theta_s - \theta_r - A_1 h_{b1}) h_{b1}^{A_2}$$
 (3)

where θ is the volumetric soil water content, h is the soil water pressure head, θ_s and θ_r are the saturation and the residual soil water content, respectively, A_1 is a curve shape parameter, A_2 represents the pore size distribution, and h_{b1} is the breakpoint in the $\theta(h)$ curve. Parameters were obtained from best fitting of the experimental data and solving simultaneously Eqns (1) and (2) for $h = h_{b1}$.

The hydraulic conductivity curve is described by

$$K(h) = K_{sat}/h^{-N_1}, \quad 0 \leq h \leq h_{b2}/$$
 (4)

$$K(h) = C_2/h^{-N_2}, \quad h \geq h_{b2}/$$
 (5)

where K is the hydraulic conductivity of the soil, K_{sat} is the saturated hydraulic conductivity, N_1 , N_2 , and C_2 are fitting parameters, and h_{b2} is the breakpoint in the $K(h)$ curve. The parameters of the $K(h)$ curve were also obtained from best fitting of the observed data.

Table 2
Total macroporosity and fraction of dead end pores, for the silty loam soil

Depth, cm	Macroporosity, cm ³ /cm ³	Fraction of dead end pores
0–15	0.0014	0.008
15–30	0.0014	0.008
30–45	0.0001	0.006
45–60	0.0001	0.001
> 60	0.0001	0.001

The model accommodates several different layers of knowledge of these parameters. If the user has measured $\theta(h)$ and $K(h)$ data from which these parameters can be derived, then these parameters are an input to the model. When not all the values are known, they are estimated by the program from input data on soil bulk density, textural classes and either soil water content at 1/3 bar or 1/10 bar. If neither 1/3 bar or 1/10 bar water contents are known, average textural class values are employed based on bulk density and the fractions of sand, silt and clay (USDA-ARS, 1995).

For the silty loam soil, the $\theta(h)$ and $K(h)$ parameters were obtained by best fitting of the $\theta(h)$ and $K(h)$ data observed in the field. A first calibration was performed by Cameira *et al.* (1997), at the same site, but for bare soil conditions, using a two-domain approach (soil matrix and macropore flow) with the same model RZWQM. The set of parameters resulting from this calibration was used to initialize the calibration of the parameters in this study. The values of the total macroporosity and fraction of dead end pores obtained by Cameira *et al.* (1997) were kept the same, and are given in Table 2. The final calibration consisted in iteratively adjusting the shape parameters of the curves to minimize the differences between observed and model simulated soil water content profiles. Equations (6) and (7) were utilized to evaluate the quality of predictions. Data for the 1997 irrigation season were utilized. Only the parameter A_2 was required to be modified relatively to the initial set of parameters during this calibration procedure which indicates that the first calibration by Cameira *et al.* (1997) was carefully performed. The calibrated parameters of the $\theta(h)$ and $K(h)$ curves adopted in this study are presented in Table 3.

In the sandy soil, because no previous studies could be used and $\theta(h)$ and $K(h)$ observed data did not adequately cover the full range of the curves, only three parameters were calibrated: the saturated hydraulic conductivity K_{sat} , the saturation water content θ_s , and the residual water content θ_r . All other soil hydraulic parameters were estimated by the model, based on those parameters and on the soil physical characteristics using the methods

Table 3
Calibrated parameters for the silty loam soil for : (a) the soil water retention curve; and (b) the hydraulic conductivity curve

(a) Soil water retention $\theta(h)$ curve					
Depth, cm	Saturation water content (θ_s), cm ³ /cm ³	Residual soil water content (θ_r), cm ³ /cm ³	Curve breakpoint (h_{b1}), cm	Curve shape parameter (A_1), cm ⁻³	Pore size distribution (A_2)
0–15	0.50	0.05	84.05	0.0031	0.40
15–30	0.40	0.05	140.63	0.0006	0.35
30–45	0.39	0.06	87.29	0.0005	0.20
45–60	0.39	0.06	45.18	0.0005	0.10
60–90	0.39	0.05	18.32	0.0006	0.03
> 90	0.39	0.05	18.32	0.0006	0.03

(b) Hydraulic conductivity $K(h)$ curve					
Depth, cm	Saturated hydraulic conductivity (K_{sat}), cm/h	Curve breakpoint (h_{b2}), cm	Curve fitting parameters		
			C_2 , cm/h	N_1	N_2
0–15	1.00	1.0	1.00	0.0	1.326
15–30	1.00	1.0	1.00	0.0	1.326
30–45	0.96	1.0	0.90	0.0	1.390
45–60	0.90	1.0	0.90	0.0	1.390
60–90	2.20	14.9	0.79	1.91	1.270
> 90	2.20	14.9	0.79	1.91	1.270

described by Ahuja and Webson (1992). No macropore flow was considered to simulate the soil water in the sandy soil. The calibration was performed using the soil water profiles observed during 1997. The calibrated parameters adopted for this soil type are shown in Table 4. Parameters estimated by the model are not given.

3.2.3. Calibration of the herbicide component

Herbicide dissipation was simulated in a lumped one-compartment model. The herbicide parameter calibrated in this study was herbicide half-life. The difference between the observed and simulated atrazine concentration in the soil profile was minimized, and the best value for atrazine half-life was found. The input herbicide parameters for the herbicide component are given in Table 5.

3.2.4. Statistical evaluation of the simulated data

Statistical parameters of goodness-of-fit were used to compare the observed and simulated soil water content and atrazine concentrations: maximum error e_{max} , and average absolute error e_{aa} . They are defined as

$$e_{max} = \max(|O_i - S_i|)_{i=1}^n \quad (6)$$

$$e_{aa} = \sum_{i=1}^n |O_i - S_i|/n \quad (7)$$

where O_i represents the observed values, S_i represents the predicted or simulated values, and n is the number of observations. For ideal prediction, the values of e_{max} and e_{aa} should be equal to zero. A least-squares linear regression analysis through the origin was used to evaluate the

Table 4
Calibrated parameters for the hydrological component, for the sandy soil

Depth, cm	Saturated hydraulic conductivity (K_{sat}), cm/h	Saturation water content (θ_s), cm ³ /cm ³	Residual soil water content (θ_r), cm ³ /cm ³
0–15	40.0	0.37	0.003
15–30	40.0	0.37	0.01
30–45	40.0	0.37	0.01
45–60	60.0	0.30	0.01
> 60	60.0	0.28	0.01

Table 5
Calibrated and other input parameters for the pesticide component

<i>Atrazine</i>		
Physiochemical properties	Molecular weight, g/mol	215.69
	Lumped half-life, days*	13 [†] ; 15 [‡]
	Temperature at which half-life is considered, °C	25
	Henry's law constant	1 × 10 ⁻⁵
	Oxygen content during anaerobic conditions, %	15
	Water solubility, µg/l	33
	Daughter product formation percentage, %	20
Equilibrium coefficients	Acid dissociation	1 × 10 ⁻⁶
	Base protonation	0.012
	Sorption constant for soil organic matter	100
	Kinetic equilibrium constant for adsorption	0.0
	Kinetic equilibrium constant for removal from soil surface	0.125

*Calibrated values for 1997. [†]For the silty loam soil. [‡]For the sandy soil.

performance of the RZWQM in predicting soil water content and atrazine concentrations in the soil profile. The coefficient of determination *r*² resulting from the regression is a measure of the degree of association between measured and simulated data. The coefficient of regression indicates the trend of predictions. For visual analysis the observed values were plotted against the simulated values and the distribution of the points around the 1:1 line was used as a qualitative indicator of model performance.

The final parameter set for each one of the three calibrations (plant growth, hydrological and pesticide component) was selected according to the best visual analysis of the data, and the best distribution of the simulated *versus* observed data around the 1:1 line, together with the regression coefficient and the coefficient

of determination relative to the regression through the origin between simulated and observed values.

4. Results and discussion

4.1. Plant development and yield

Table 6 presents the observed and simulated yields, for both soil types, in 1996 and 1997. Model predictions for 1997 (calibration year) were very close to the observed values. However, in 1996, for the sandy soil plot, the difference between the observed and simulated yield was higher (23.5%). During the summer of 1996, this plot had an infestation of mites which produced a significant decrease in green mass. Infestations cannot be simulated by the RZWQM. Therefore, the model was not able to accurately predict yield in those conditions.

Figure 1 shows the simulated and observed leaf area index (LAI) and above ground biomass, in the sandy soil, for the calibration year (1997). Results were similar for the silty loam soil during the same growing season. The predicted above ground biomass was close to the observed values, for both soil types. Regarding LAI, model predictions for the sandy soil were good. However, for the silty loam soil, the model tends to overpredict LAI during the growing season.

The statistical parameters for the comparison between the observed and the simulated plant growth and yield of the maize crop for both soils, during 1997, are presented in Table 7. These results indicate that the RZWQM is effective in simulating plant growth and yield for the maize crop.

4.2. Soil water content

4.2.1. Silty loam soil

Figure 2 shows the simulated and observed soil water content in the soil profile in the silty loam soil, for the calibration year (1997), before and after the irrigation

Table 6
Observed and simulated yields (d.b.) for the silty loam and sandy soils, for 1996 and 1997 (calibration year)

<i>Soil type</i>	<i>Year</i>	<i>Crop</i>	<i>Observed yield, t/ha</i>	<i>Simulated yield, t/ha</i>	<i>Difference, t/ha</i>	<i>Difference, %</i>
Silty loam soil	1996	Grain	12.3	10.1	- 2.2	17.9
	1997	Forage (d.b.)	20.7	20.1	- 0.6	2.9
Sandy soil	1996	Forage (d.b.)	16.2*	20.0	+ 3.8*	23.5
	1997	Grain	8.6	8.4	- 0.2	2.3

*In 1996, an estimated 20% of the yield was lost, in the sandy soil plot, due to an infestation of mites, therefore the production could have been around 20 t/ha (d.b.).

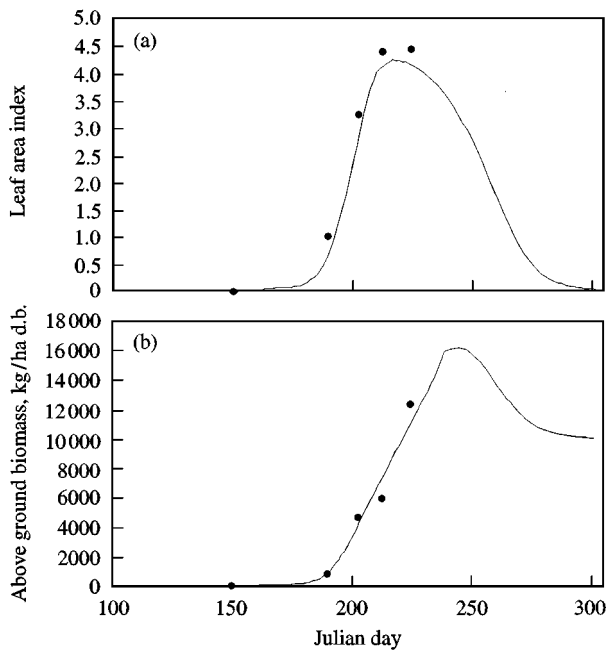


Fig. 1. Simulated (—) and observed (●) crop parameters for the sandy soil for the calibration year (1997): (a) leaf area index; (b) above ground biomass

events. A visual analysis of Fig. 2 shows that (except for the soil water in the profile after the second irrigation) the model is able to predict the soil water content profiles. In fact, for irrigations 3 and 4, the simulated values fall in the range of the observed soil water content, before and after the irrigation events and are not far from those intervals for irrigation 5. However, the model underpredicts the soil water content after the second irrigation event. Soil variability and non-uniformity of irrigation may have contributed to this result, as well as any variation in the inflow discharge to the basin leading to an overestimation of the applied water depth during this event.

Figure 3 shows the distribution of the observed *versus* simulated values of soil water content around the 1:1 line. Data are equally distributed along the 1:1 line for the calibration year, despite it is somewhat scattered for the higher θ values [Fig. 3(a)].

For the evaluation year (1996), the model predictions show a tendency to underpredict the soil water content [Fig. 3(b)]. The fact that 1995/1996 was very wet, with fields flooded during part of the wintertime, meant that the water table was very high from the beginning of the irrigation season. Therefore, the upward fluxes from the water table largely contributed to the soil water content, which cannot properly be accounted for in the model and causes underprediction of the soil water content in these conditions. To evaluate the performance of the model, the maximum error and the average absolute error were calculated. A linear regression through the origin was also performed. These results are shown in Table 8, and confirm the analysis above. The average absolute errors are low and the slope of the linear regression is not far from 1. Overall, model predictions of soil water content show the capability of the model to predict soil water content in this soil.

4.2.2. Sandy soil

Figure 4 shows the simulated and observed soil water content in the soil profile, for the sandy soil, for the calibration year 1997. A visual analysis of this figure shows that an acceptable agreement between simulated and observed values was obtained, before and after the irrigation events. The observed trends of θ variation with depth are clearly followed by the simulation. Figure 5(a) presents the comparison between the observed and simulated soil water content and the 1:1 line. Observed data in Fig. 5(a) show great variability, sometimes with very large range of variation. This variation is due to both the soil variability and the non-uniformity of irrigation, which was often affected by the wind. These

Table 7

Summary of the statistical analysis to evaluate the prediction of plant development parameters, in the silty loam and the sandy soils for 1997 (calibration year)

Statistical parameter	Maximum error	Average absolute error	Coefficient of determination*	Coefficient of regression*
Silty loam soil				
Leaf area index	1.40	0.65	0.98	1.25
Dry mass, kg/ha	1494	823	0.93	0.98
Sandy soil				
Leaf area index	0.34	0.23	1.00	0.93
Dry mass, kg/ha	1488	671	0.96	0.96

*The regression through the origin between simulated and observed data.

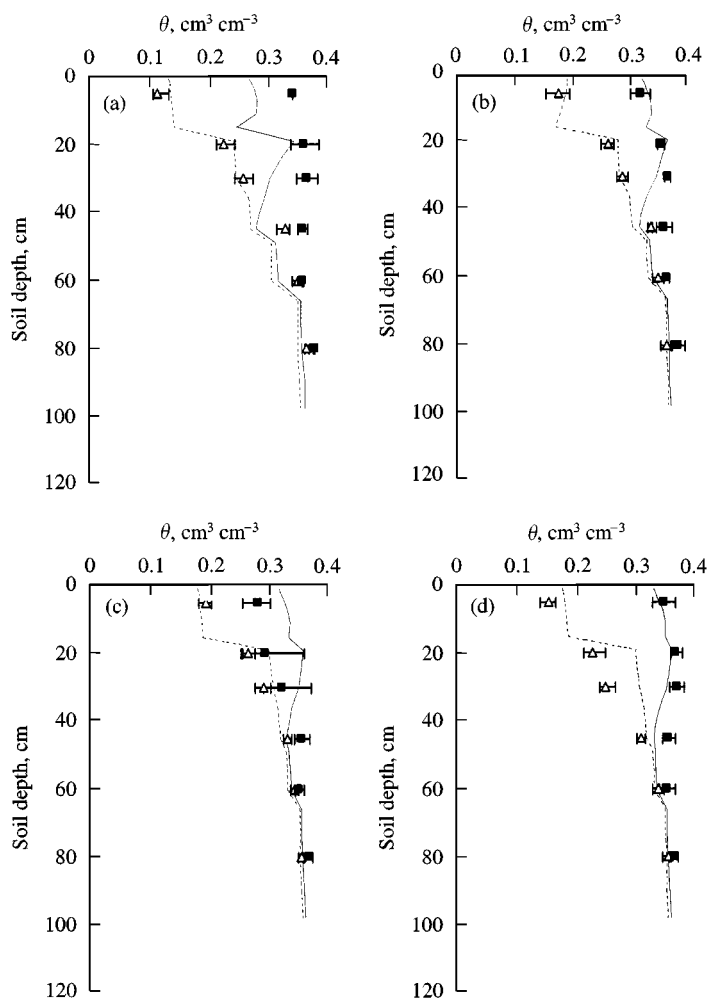


Fig. 2. Simulated and observed soil water content in the soil profile in the silty loam soil for the calibration year (1997). (a) Δ , before and $\blacksquare$, after irrigation 2; (b) Δ , before and $\blacksquare$, after irrigation 3; (c) Δ , before and $\blacksquare$, after irrigation 4; (d) Δ , before and $\blacksquare$, after irrigation 5; θ , volumetric water content; (---) simulated before irrigation; (—) simulated after irrigation

conditions make model simulation more difficult. The summary of the statistical analysis using these data is presented in Table 8. Results indicate that model predictions of the soil water content are appropriate.

For the evaluation year (1996), simulated and observed soil water content is not shown. However, the comparison between observed and simulated values is presented in Fig. 5(b). Model predictions are not as good as for the

Table 8
Summary of the statistical analysis to evaluate the prediction of soil moisture content in the soil profile

Statistical parameter	Maximum error, cm^3/cm^3	Average absolute error, cm^3/cm^3	Coefficient of determination*	Coefficient of regression*
Silty loam soil				
1997	0.11	0.03	0.62	0.97
1996	0.12	0.04	0.70	0.90
Sandy soil				
1997	0.08	0.02	0.25	0.95
1996	0.10	0.04	0.24	0.79

*The regression through the origin between simulated and observed data.

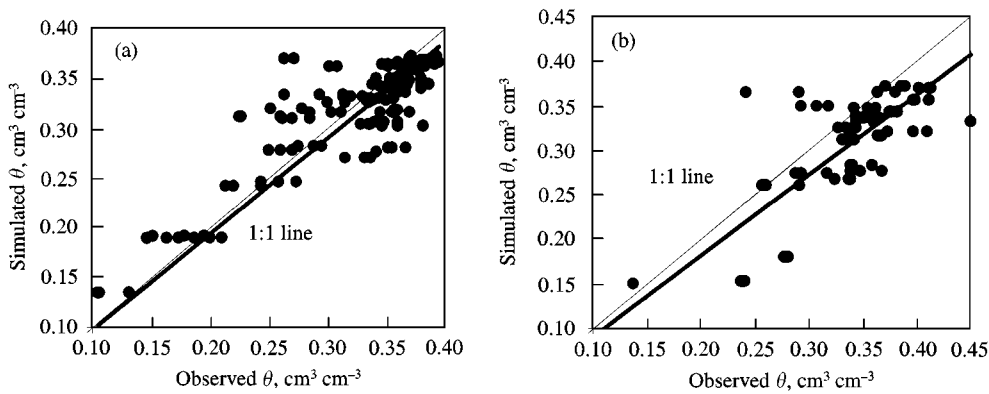


Fig. 3. Comparison between the observed versus simulated soil water content θ and the 1 : 1 line for the silty loam soil: (a) during the calibration year (1997); (b) during 1996; (—) regression through the origin

calibration year. For 1996, surface irrigation was applied, which is a method not appropriate for these soils and causing great variability. Therefore, soil water observations were affected by this variability, that could not be taken into consideration in the simulation.

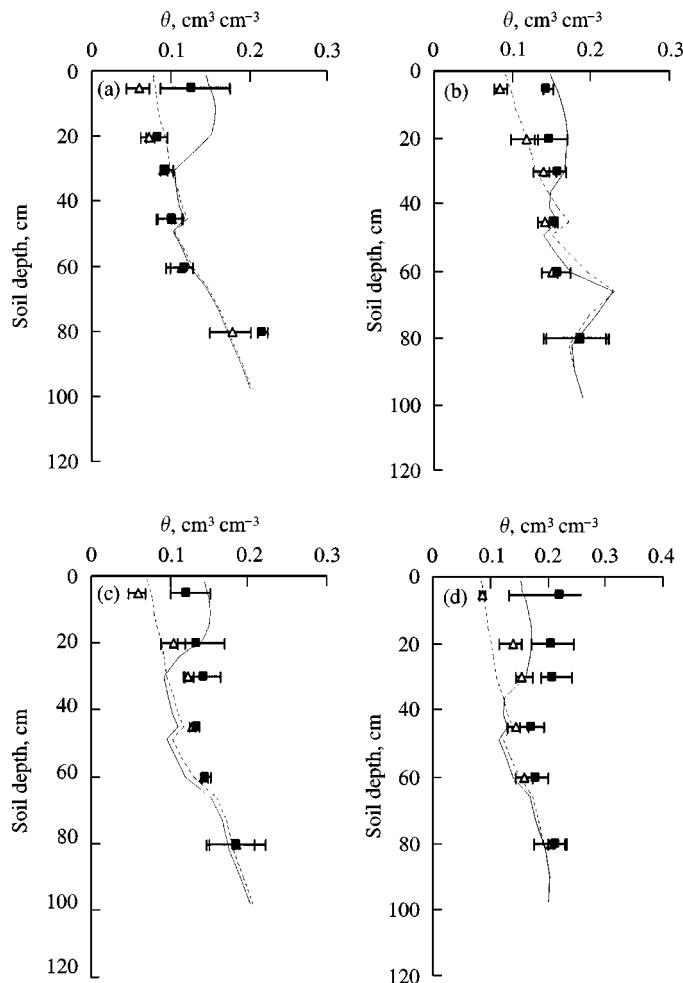


Fig. 4. Simulated and observed soil moisture content in the soil profile in the sandy soil for the calibration year (1997): (a) 9 July (Δ , before irrigation; $\blacksquare$, after irrigation); (b) 22 July (Δ , before irrigation; $\blacksquare$, after irrigation); (c) 20 August (Δ , before irrigation; $\blacksquare$, after irrigation); (d) 9 September (Δ , before irrigation; $\blacksquare$, after irrigation); θ , volumetric water content; (----) simulated before irrigation; (—) simulated after irrigation

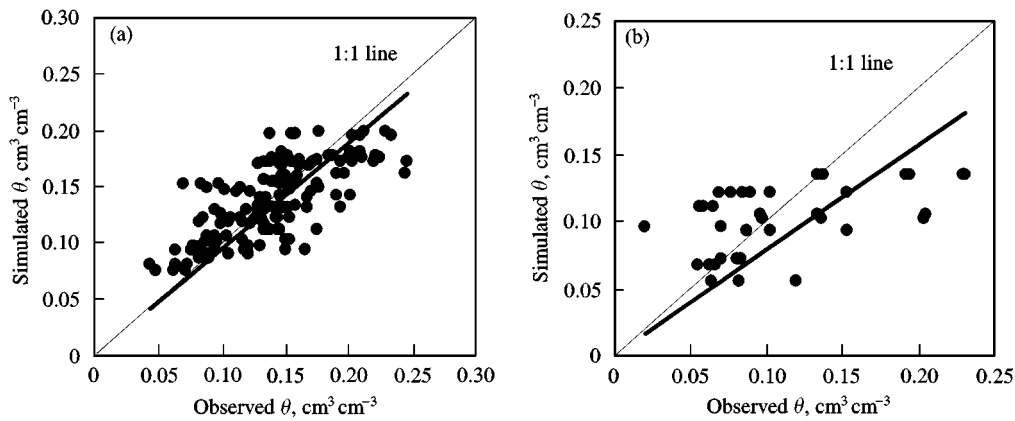


Fig. 5. Overall comparison between the observed versus simulated soil water content θ , and the 1 : 1 line for the sandy soil: (a) during the calibration year (1997); (b) during 1996; (—) regression through the origin

For both soil types, the model predictions are in agreement with the observed values, showing that RZWQM can be used, when properly calibrated, for Portuguese conditions. In the calibration year, for most of the irrigations, the model performance was good. However, the model needs to be improved to deal with shallow water tables, to better consider the resulting upward fluxes into the root zone.

4.3. Atrazine in soil

4.3.1. Silty loam soil

During the calibration year (1997), the calibrated half-life was 13 days, which was smaller than the calculated half-life using the observed values (23 days). This difference may be due to the fact that the lumped herbicide dissipation by a one-compartment model is not accurate enough to describe the herbicide processes in soil.

Simulated atrazine concentrations after herbicide application were very close to the observed concentrations (data not shown). For the other three sampling dates, model predictions of atrazine concentration in the top 20 cm were not so good. However, the model correctly predicted the depth of atrazine penetration in the soil profile. Also, atrazine concentrations in the soil after 22 July were very small, therefore, the absolute error for these dates is small (less than 0.1 p.p.m.).

Figure 6 presents the comparison between the observed and simulated atrazine concentrations and the 1 : 1 line. A summary of the statistical analysis to evaluate the prediction of atrazine concentration in the soil profile is given in Table 9. Results in this table and in Fig. 6(a) indicate that the model overpredicts atrazine concentrations in 1997. However, the average absolute error is small (0.035 p.p.m.).

Figure 7 shows the simulated and observed atrazine concentrations in the soil profile for the evaluation year,

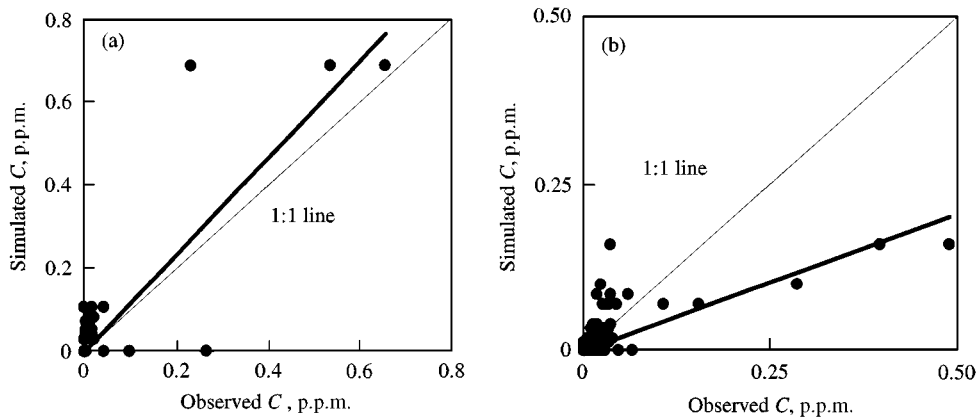


Fig. 6. Overall comparison between the observed versus simulated atrazine concentrations C , and the 1 : 1 line for the silty soil: (a) during the calibration year (1997); (b) during 1996; (—) regression through the origin

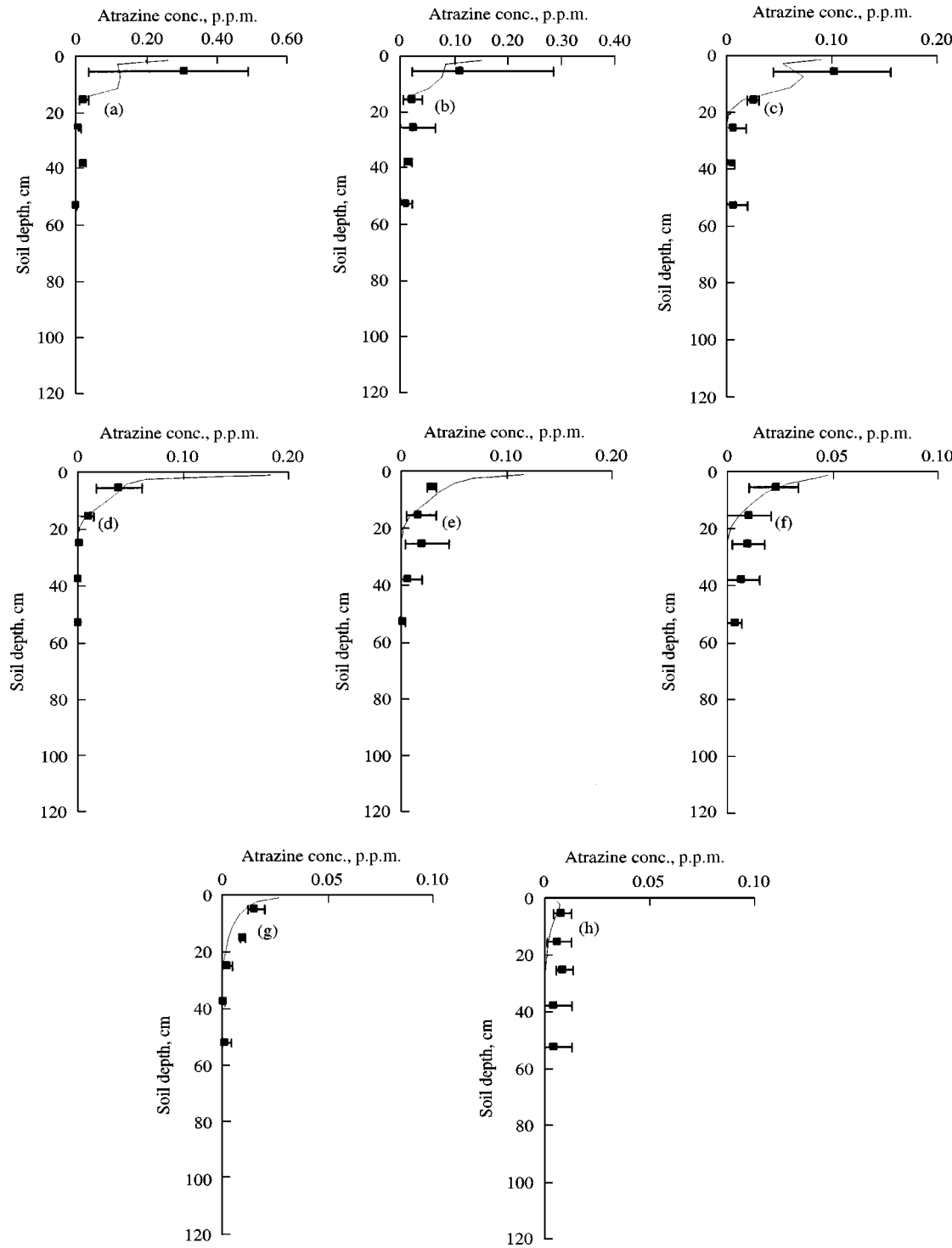


Fig. 7. Simulated and observed atrazine concentrations in the soil profile in the silty loam soil for 1996: (a) 27 May; (b) 1 July — before irrigation 1; (c) 4 July — after irrigation 1; (d) 22 July — before irrigation 2; (e) 26 July — after irrigation 2; (f) 14 August — after irrigation 3; (g) 3 September — after irrigation 4; (h) 25 September; —, simulated

1996. A visual analysis of this figure shows that simulated profiles follow the trend of the observed ones for all the sampling dates. Simulated atrazine concentrations were in the range of the observed values. The comparison

between the observed and simulated atrazine concentrations in Fig. 6(b) and the results in Table 9 indicate that the model underpredicts atrazine concentrations for 1996. As for the soil water simulation, these results may

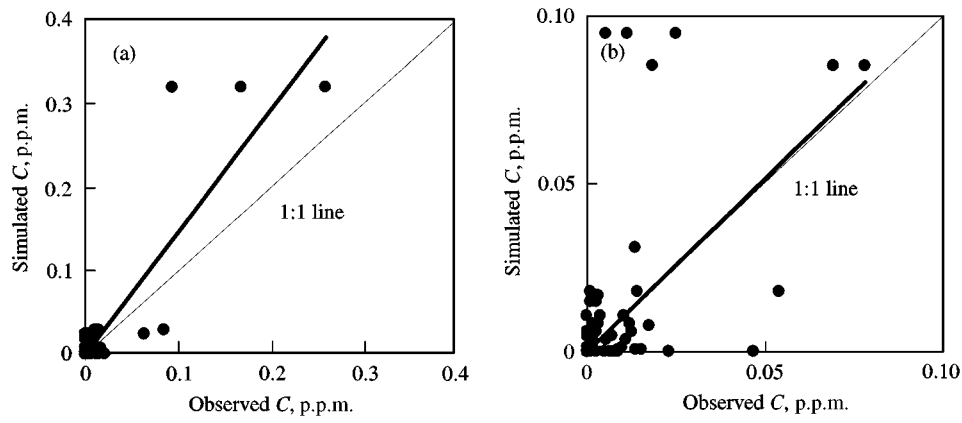


Fig. 8. Overall comparison between the observed versus simulated atrazine concentrations *C*, and the 1 : 1 line for the sandy soil: (a) during the calibration year (1997); (b) during 1996; (—) regression through the origin

have been influenced by the high level of the perched water table during this year. Because the model could not fully account for the upward water fluxes, it was also not able to consider that these fluxes were opposing the downward transport of atrazine. Results indicate that the model is able to simulate atrazine transport in the silty loam soil.

4.3.2. Sandy soil

For the calibration year, the calibrated half-life was 15 days. As for the silty loam soil, the calibrated half-life was smaller than the calculated half-life with the observed values (19 days). However, the difference is small. Simulated and observed atrazine concentrations in the soil profile (data not shown) follow a similar trend. The comparison between the observed and the simulated atrazine concentrations and the 1 : 1 line is presented in Fig. 8. Table 9 gives the summary of the statistical analysis to evaluate the prediction of atrazine concentrations in the soil profile. In 1997, the slope of the linear regression is higher than 1 showing that the model overpredicts atrazine concentration. However, this value is strongly

influenced by the differences between observed and simulated values in the layer 0–10 cm on 13 May. The overall average absolute error is 0.012 p.p.m.

For the evaluation year, 1996, the simulated and observed atrazine concentrations in the soil profile are presented in Fig. 9. Observing the four successive profiles in this figure, one can conclude that the RZWQM is able to simulate the atrazine transport for deeper layers. The simulated and observed atrazine concentrations were generally within the same range. Fig. 8(b) shows the distribution of the atrazine concentration values along the 1 : 1 line. It confirms that the RZWQM is effective in simulating atrazine concentrations for the sandy soils. This fact is confirmed by the statistical parameters in Table 9. However, the dispersion along the 1 : 1 line is high, while the average absolute error is 0.013 p.p.m.

Results indicate that model predictions on atrazine concentration in the soil were in agreement with the observed values for both soils (silty loam and sandy soil). Nevertheless, for the prevailing experimental conditions, the model does a better job in simulating atrazine concentrations in the sandy soil. Results also indicate that

Table 9
Summary of the statistical analysis to evaluate the prediction of atrazine concentration in the soil profile

Statistical parameter	Maximum error p.p.m.	Average absolute p.p.m.	Coefficient of determination*	Coefficient of regression*
Silty loam soil				
1997	0.457	0.035	0.73	1.16
1996	0.333	0.018	0.45	0.41
Sandy soil				
1997	0.225	0.012	0.78	1.47
1996	0.089	0.013	0.22	1.03

*The regression through the origin between simulated and observed data.

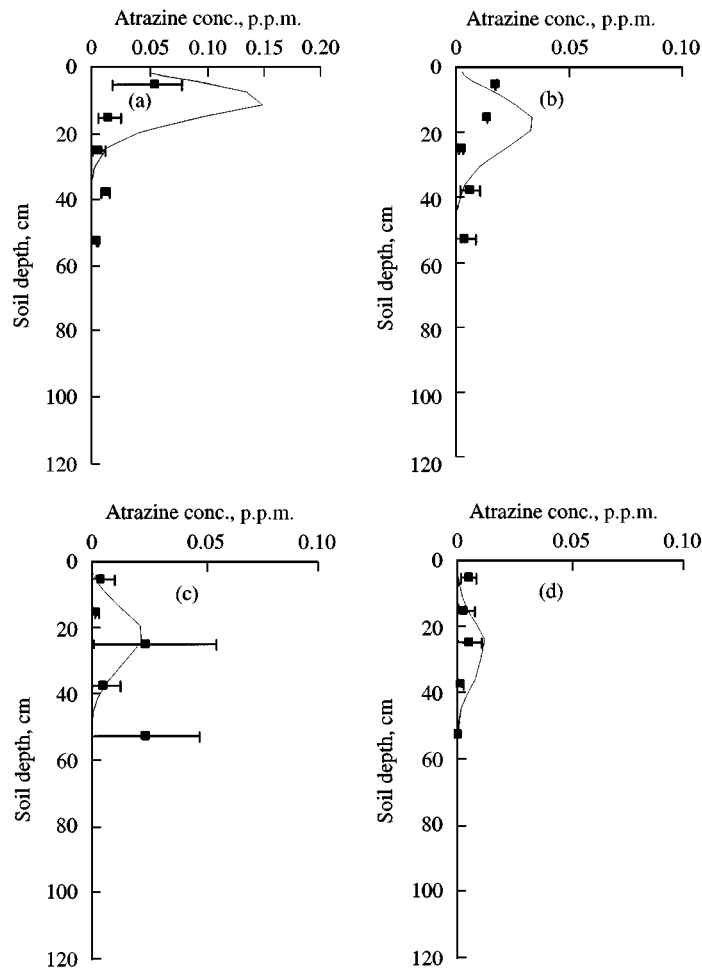


Fig. 9. Simulated and observed atrazine concentrations in the soil profile in the sandy soil for 1996: (a) 27 May; (b) 10 July; (c) 30 July; (d) 29 August; —, simulated

the model performance for atrazine simulation relates to the conditions set for simulation of the soil water contents. In this study, it was also shown that there was no evidence of atrazine transport in the silty loam soil while atrazine leaching had occurred in the sandy soil due to percolation produced by overirrigation.

5. Conclusions

This study resulted in the following conclusions.

The root zone water quality model (RZWQM) is effective in simulating plant growth parameters. However, when predicting yields, and because the RZWQM does not consider crop infestations, this was found to be a source of error in estimating yields for 1 year.

For both soil types, model predictions of the soil water content profiles were in agreement with the observed values. In the calibration year, for most of the irrigations, the model performance was good. However, the model could not accurately account for the contributions from

a shallow water table. Predictions were also affected by the soil variability and the non-uniformity of irrigation.

Results indicate that model predictions on atrazine concentrations in the soil were in acceptable agreement with the observed values for both soil types. The model has shown that atrazine transport to deeper layers out of the root zone, was relatively important in the sandy soil while such behavior was not observed in the silty loam soil.

This study showed that the RZWQM can be used, when properly calibrated, to predict soil water content and atrazine movement through the soil profile for Portuguese conditions.

Acknowledgements

Funding for this project was provided by the Centro de Estudos de Engenharia Rural (CEER), ISA-UTL, and the PRAXIS XXI program, Portugal.

References

- Ahuja L R** (1983). Modeling infiltration into crusted soils by the Green-Ampt approach. *Soil Science Society of America Journal*, **47**, 412–418
- Ahuja L R; DeCoursey D G; Barnes B B; Rojas K W** (1993). Characterization of macropore transport studied with the ARS root zone water quality model. *Transactions of the ASAE*, **36**(2), 369–380
- Ahuja L R; Ma L; Hanson J D** (1998). Application of the Root Zone Water Quality Model for environmental-water management in agricultural systems. In: *Water and the Environment: Innovation Issues in Irrigation and Drainage* (Pereira L S; Gowing J eds), pp 3–11. E&FN Spon, London
- Ahuja L R; Ma Q L; Rojas K W; Boesten J J; Farahani H J** (1996). A field test of root zone water quality model—pesticide and bromide behavior. *Pesticide Science*, **48**, 101–108
- Ahuja L R; Webson** (1992). Water chemical and heat transport in soil matrix and macropores. In: *Root Zone Water Quality Model. Technical Documentation*, USDA/ARS, Fort Collins, CO
- Azevedo A S; Kanwar R S; Singh P; Ahuja L; Pereira L S** (1997a). Simulating atrazine transport through soil profile using RZWQM for an Iowa soil. *Journal of Environmental Quality*, **26**(1), 153–164
- Azevedo A S; Singh P; Kanwar R S; Ahuja L R** (1997b). Simulating nitrogen management effects on subsurface drainage water quality. *Agricultural Systems*, **55**(4), 481–501
- Azevedo A S; Kanwar R S; Pereira L S** (2000). Atrazine transport in irrigated heavy and coarse textured soils. Part I: field studies. *Journal of Agricultural Engineering Research*, doi: 10.1006/jaer.2000.0543
- Brooks R H; Corey A T** (1964). Hydraulic properties of porous media. *Hydrology Paper 3*, pp 1–15, Colorado State University, Fort Collins, CO
- Cameira M R; Fernando R M; Pereira L S** (1997). Modeling infiltration and redistribution in a plowed soil with a two domain flow model. In: *Proceedings of the International Conference on “Water management, salinity, and pollution control towards sustainable irrigation in the mediterranean region”*, Vol. II, pp 287–306. IAM, Bari
- Cameira M R; Sousa P L; Farahani H J; Ahuja L R; Pereira L S** (1998). Evaluation of the RZWQM for the simulation of water and nitrate movement in level basin, fertilized maize. *Journal of Agricultural Engineering Research*, **69**, 331–341
- Celia M A; Bouloutaas E T; Zarba R L** (1990). A general mass-conservative numerical solution for the unsaturated flow equation. *Water Resources Research*, **26**, 1483–1496
- Farahani H J; Ahuja L R; Peterson G A; Sherrod L A; Mrabet R** (1995). Root zone water quality model evaluation of dry-land/no-till crop production in Eastern Colorado. In: *Proceedings of the International Symposium on Water Quality Modeling* (Heatwole C eds). ASAE, St. Joseph, MI
- Funari E; Donati L; Sandroni D; Vighi M** (1995). Pesticide levels in groundwater: value and limitations of monitoring. In: *Pesticide Risk in Groundwater* (Vighi M; Funari E eds), pp 3–43. Lewis Publishers, FL
- Hallberg G R** (1989). Pesticide pollution of groundwater in the humid United States. *Agricultural Ecosystems Environment*, **26**, 299–367
- Johnsen K E; Liu H H; Dane J H; Ahuja L R; Workman S R** (1995). Simulating fluctuating water tables and tile drainage with a modified root zone water quality model and a new model WAFLOWM. *Transactions of the ASAE*, **38**(1), 75–83
- Ma L; Selim H M** (1997). Physical non-equilibrium modelling approaches to solute transport in soils. *Advances in Agronomy*, **58**, 95–150
- Nash R G** (1988). Models for estimating pesticides dissipation from soil and vapor decline in air. *Chemosphere*, **18**, 2374–2381
- Nokes S E; Landa F M; Hanson J D** (1995). Validation of the root zone water quality (RZWQM) from a crop growth perspective. In: *Proceedings of the International Symposium on Water Quality Modelling* (Heatwole C eds). ASAE, St. Joseph, MI
- Pereira L S; Cameira M R** (1999). Unsaturated zone leaching models: a user's view. *ICID Journal*, **48**(2), 1–9
- Singh P; Johnsen K E; Kanwar R S; Ahuja L R** (1995). Simulating the effects of soil types on subsurface drain flows by using RZWQM. In: *Proceedings of the International Symposium on Water Quality Modelling* (Heatwole C eds). ASAE, St. Joseph, MI
- Singh P; Kanwar R S; Johnsen K E; Ahuja L R** (1996). Calibration and evaluation of subsurface drainage component of RZWQM V2.5. *Journal of Environmental Quality*, **25**, 56–63
- Stoltenberg D; Pope R; Most J** (1990). Atrazine management rules for Iowa. *Iowa State University Cooperation Extension Bulletin*, **1390**, 1–6
- USDA-ARS** (1992). Root zone water quality model (RZWQM) V. 1.0. Technical Documentation. GPSR Technical Report No.2, USDA-ARS Great Plains Systems Research Unit. Ft. Collins, CO
- USDA-ARS** (1995). Root zone water quality model (RZWQM) V. 3.0. User's Manual. GPSR Technical Report No.5. USDA-ARS Great Plains Systems Research Unit. Ft. Collins, CO
- Wagenet R J; Rao P S C** (1990) Modelling pesticide fate in soils. In: *Pesticides in the Soil Environment: Processes, Impacts and Modelling*, SSSA Books series 2 (Cheng H H eds). SSSA